LEVEL 3 — NumPy Indexing and Slicing

Indexing means selecting specific elements. Slicing means selecting a range of elements.

PART A — 1D ARRAY

```
import numpy as np

arr = np.array([10, 20, 30, 40, 50])
```

Index positions:

```
Index:   0   1   2   3   4
Value:  10  20  30  40  50
```

1. Positive Indexing

```
print(arr[2])
```

Output: 30

Indexing starts from 0, so index 2 contains 30.

2. Negative Indexing

```
print(arr[-2])
```

Output: 40

Negative indexing starts from the end. -1 is the last element.

3. First Element

```
print(arr[0])
```

Output: 10

4. Last Element

```
print(arr[-1])
```

Output: 50

5. Multiple Elements

```
print(arr[[0, 2, 4]])
```

Output: [10 30 50]

This selects elements at indexes 0, 2, and 4.

6. Slicing

```
print(arr[1:4])
```

Output: [20 30 40]

Start is included, stop is excluded: indexes 1, 2, 3.

7. Step Slicing

```
print(arr[0:5:2])
```

Output: [10 30 50]

Syntax: arr[start:stop:step].

8. Reverse Array

```
print(arr[::-1])
```

Output: [50 40 30 20 10]

A step of -1 moves from the end to the beginning.

PART B — 2D ARRAY

```
arr2 = np.array([[10, 20, 30],
                 [40, 50, 60],
                 [70, 80, 90]])
```

Rows and columns:

```
        Col 0  Col 1  Col 2
Row 0     10     20     30
Row 1     40     50     60
Row 2     70     80     90
```

1. Row Selection

```
print(arr2[1])
```

Output: [40 50 60]

2. Column Selection

```
print(arr2[:, 1])
```

Output: [20 50 80]

The colon selects all rows; 1 selects column 1.

3. Specific Element

```
print(arr2[1, 2])
```

Output: 60

Syntax: arr2[row, column].

4. Multiple Rows

```
print(arr2[[0, 2]])
```

Output: [[10 20 30] [70 80 90]]

5. Multiple Columns

```
print(arr2[:, [0, 2]])
```

Output: [[10 30] [40 60] [70 90]]

6. Row Slicing

```
print(arr2[0:2, :])
```

Output: [[10 20 30] [40 50 60]]

Selects rows 0 and 1, and all columns.

7. Column Slicing

```
print(arr2[:, 0:2])
```

Output: [[10 20] [40 50] [70 80]]

8. Matrix Slicing

```
print(arr2[0:2, 1:3])
```

Output: [[20 30] [50 60]]

Syntax: arr2[row_start:row_stop, col_start:col_stop].

PART C — 3D ARRAY

```
arr3 = np.array([
    [[1, 2, 3], [4, 5, 6]],
    [[7, 8, 9], [10, 11, 12]]
])
```

This is a 3D array with 2 layers, 2 rows per layer, and 3 columns per row.

1. Accessing Dimensions

```
print(arr3.ndim)
print(arr3.shape)
```

Output: 3 (2, 2, 3)

Meaning: 2 layers × 2 rows × 3 columns.

2. Selecting Layers

```
print(arr3[0])
```

Output: [[1 2 3] [4 5 6]]

arr3[0] selects the first layer.

3. Selecting Rows

```
print(arr3[:, 1, :])
```

Output: [[4 5 6] [10 11 12]]

Selects row 1 from every layer.

4. Selecting Columns

```
print(arr3[:, :, 2])
```

Output: [[3 6] [9 12]]

Selects column 2 from every layer and every row.

5. Advanced Slicing

```
print(arr3[:, 0:2, 1:3])
```

Output: [[[2 3] [5 6]] [[8 9] [11 12]]]

This selects all layers, rows 0–1, and columns 1–2.

QUICK REVISION

```
1D:  arr[index]
     arr[start:stop]
     arr[start:stop:step]
     arr[::-1]                 → reverse

2D:  arr[row, column]
     arr[row, :]
     arr[:, column]
     arr[row_start:row_stop, col_start:col_stop]

3D:  arr[layer, row, column]
     : means select all values along that dimension
```